🌱 Environment — Master File (Species + Others) · SSC CGL

> **The single consolidated Environment resource** — ecology basics, biodiversity & species, protected areas, pollution, climate & conventions, with PYQ question-types, curated & new questions. Per the PYQ Blueprint.

> ⚠️ **Current figures move** (Ramsar count, censuses, rankings) — verify before the exam. Static concepts (conventions, food chains, pollutants) rarely change.

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1. How SSC Asks Environment (PYQ Types)

Type	Example stem	Prep
Convention → subject	"Ramsar Convention is about —"	Conventions (§6)
Species → status/park	"Asiatic lion is found in —"	Species & parks (§3–4)
Pollutant → effect	"Which gas causes acid rain?"	Pollution (§5)
Term/definition	"The base of a food chain is —"	Ecology (§2)
Project → target	"Project Tiger was launched in —"	Conservation (§4)

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2. Ecology & Ecosystem Basics

- **Ecosystem** = biotic (living) + abiotic (non-living) components interacting.
- **Food chain:** Producers (green plants — autotrophs) → Herbivores (primary consumers) → Carnivores. **~10% energy** transfers to the next trophic level (10% law).
- **Food web** = interconnected food chains. **Ecological pyramid** (energy pyramid always upright).
- **Producers** form the base; **decomposers** (bacteria, fungi) recycle nutrients.
- **Biosphere Reserve → National Park → Wildlife Sanctuary** (decreasing protection strictness varies; NP most strict on human activity).

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3. Biodiversity & Species

- **Biodiversity** = variety of life; India is a **mega-diverse** country (2 biodiversity hotspots wholly/partly: Western Ghats, Eastern Himalayas, Indo-Burma, Sundaland).
- **IUCN Red List categories:** Extinct → Extinct in Wild → **Critically Endangered** → **Endangered** → **Vulnerable** → Near Threatened → Least Concern.
- **Flagship species:** Tiger (national animal), Asiatic lion (Gir), One-horned rhino (Kaziranga), Snow leopard (Himalayas), Great Indian Bustard (critically endangered), Gangetic dolphin (national aquatic animal).
- **National symbols:** National animal — **Tiger** · National bird — **Peacock** · National aquatic animal — **Gangetic Dolphin** · National heritage animal — **Elephant** · National tree — Banyan · National flower — Lotus.

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4. Protected Areas & Conservation Projects

- **Project Tiger — 1973** · **Project Elephant — 1992** · **Project Cheetah** (reintroduction, Kuno NP, MP).
- **Ramsar Convention (1971, Iran):** protects **wetlands of international importance**.
- **National Parks:** Jim Corbett (1936, first), Kaziranga (rhino), Gir (Asiatic lion), Sundarbans (tiger, mangrove), Hemis (largest, snow leopard), Kanha/Bandhavgarh (MP tigers).
- **Biosphere Reserves:** Nilgiri (first, 1986), Nanda Devi, Sundarbans, Gulf of Mannar.
- **CITES** — controls international trade in endangered species. **Wildlife Protection Act, 1972** — legal backbone in India.

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5. Pollution

- **Air pollutants:** CO, SO_2, NO_2, particulate matter (PM2.5/PM10), ground-level ozone.
- **Acid rain** — caused by SO_2 & NO_x (→ sulphuric/nitric acid).
- **Greenhouse gases:** CO_2, CH_4 (methane), N_2O, water vapour, CFCs → **global warming**.
- **Ozone depletion** — caused by **CFCs** (Montreal Protocol phases them out).
- **Water pollution:** BOD (Biological Oxygen Demand) — high BOD = more pollution. Eutrophication = nutrient over-enrichment → algal bloom.
- **National Air Quality Index (AQI)** categories: Good → Satisfactory → Moderate → Poor → Very Poor → Severe.

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6. Climate Change & Global Conventions

Convention / Body	About
Ramsar (1971)	Wetlands
CITES (1973)	Trade in endangered species
Montreal Protocol (1987)	Ozone-depleting substances (CFCs)
UNFCCC (1992, Rio)	Climate change framework
Kyoto Protocol (1997)	Emission-reduction targets
Paris Agreement (2015)	Limit warming to <2 °C (aim 1.5 °C)
CBD (1992)	Convention on Biological Diversity
IPCC	UN climate science body

7. Current Environment (verify before exam)

- India's Ramsar sites crossed **100** (Surha Tal, UP — 100th, 2026).
- **Asiatic lion census (2025):** ~891 (Gujarat).
- **Project Cheetah:** Botswana handed 8 cheetahs to India (2025).
- India **9th** globally in forest area; UNESCO biosphere reserves total **13** (Cold Desert added).

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8. Curated PYQ-Pattern Questions

Q1. The Ramsar Convention deals with — (a) forests (b) **wetlands** (c) deserts (d) oceans

Q2. Project Tiger was launched in — (a) 1972 (b) **1973** (c) 1992 (d) 1985

Q3. National aquatic animal of India — (a) Whale (b) **Gangetic Dolphin** (c) Turtle (d) Crocodile

Q4. Acid rain is caused mainly by — (a) CO_2 & O_2 (b) **SO_2 & NO_x** (c) CH_4 (d) CFCs

Q5. Ozone layer depletion is caused by — (a) CO_2 (b) SO_2 (c) **CFCs** (d) methane

Q6. The Montreal Protocol is related to — (a) climate (b) **ozone-depleting substances** (c) wetlands (d) trade

Q7. In a food chain, the base is formed by — (a) carnivores (b) herbivores (c) **producers** (d) decomposers

Q8. Energy transfer to the next trophic level is about — (a) 1% (b) **10%** (c) 50% (d) 90%

Q9. The Paris Agreement aims to limit warming to below — (a) 1 °C (b) 1.5 °C (c) **2 °C** (d) 3 °C

Q10. Project Elephant was launched in — (a) 1973 (b) **1992** (c) 1985 (d) 2000

Q11. India's national heritage animal — (a) Tiger (b) Lion (c) **Elephant** (d) Rhino

Q12. The first Biosphere Reserve of India — (a) Nanda Devi (b) **Nilgiri** (c) Sundarbans (d) Gulf of Mannar

Q13. High BOD in water indicates — (a) clean water (b) **more pollution** (c) less oxygen demand (d) low bacteria

Q14. CITES controls — (a) wetlands (b) **trade in endangered species** (c) emissions (d) forests

Q15. The Wildlife Protection Act was passed in — (a) 1971 (b) **1972** (c) 1980 (d) 1986

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9. Freshly-Created Practice Questions

New questions in the SSC pattern; answer key at the end.

N1. Eutrophication in a water body is caused by —

(a) oxygen enrichment (b) nutrient over-enrichment (c) acidification (d) desalination

N2. Which UN body assesses the science of climate change?

(a) UNEP (b) IPCC (c) UNFCCC (d) WMO

N3. The Great Indian Bustard is classified as —

(a) Least Concern (b) Vulnerable (c) Critically Endangered (d) Extinct

N4. The Kyoto Protocol (1997) is associated with —

(a) wetlands (b) emission-reduction targets (c) ozone (d) biodiversity

N5. Kuno National Park, associated with Project Cheetah, is in —

(a) Rajasthan (b) Madhya Pradesh (c) Gujarat (d) Maharashtra

N6. Which gas is the most significant greenhouse gas by volume?

(a) Methane (b) CO_2 (c) N_2O (d) CFC

N7. The 10% law of energy transfer was given by —

(a) Odum (b) Lindeman (c) Tansley (d) Darwin

N8. The Convention on Biological Diversity was adopted at the —

(a) Stockholm Conference (b) Rio Earth Summit (1992) (c) Paris (2015) (d) Kyoto (1997)

N9. India's national aquatic animal is the —

(a) Olive Ridley turtle (b) Gangetic dolphin (c) Blue whale (d) Gharial

N10. The AQI category just above "Poor" is —

(a) Moderate (b) Severe (c) Very Poor (d) Satisfactory

Answer Key (Section 9):

N1-(b) nutrient over-enrichment · N2-(b) IPCC · N3-(c) Critically Endangered · N4-(b) emission-reduction targets · N5-(b) Madhya Pradesh · N6-(b) CO_2 · N7-(b) Lindeman · N8-(b) Rio Earth Summit · N9-(b) Gangetic dolphin · N10-(c) Very Poor

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10. Rapid Revision Sheet

- Food chain: producers → herbivores → carnivores; ~10% energy transfer (Lindeman)
- National symbols: animal Tiger · bird Peacock · aquatic Gangetic Dolphin · heritage Elephant
- Project Tiger 1973 · Project Elephant 1992 · Project Cheetah (Kuno, MP) · WPA 1972
- Conventions: Ramsar (wetlands) · CITES (species trade) · Montreal (ozone/CFC) · UNFCCC/Kyoto/Paris (climate) · CBD (biodiversity)
- Acid rain → SO_2 & NO_x · Ozone hole → CFCs · GHGs → CO_2/CH_4/N_2O · high BOD → more pollution
- First Biosphere Reserve → Nilgiri · IUCN: CR → EN → VU → NT → LC
- Paris Agreement → limit warming <2 °C (aim 1.5) · IPCC → climate science body
- (Current, verify) 100+ Ramsar sites · ~891 Asiatic lions · 13 biosphere reserves

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⚠ Current figures change — verify before exam. Back to Static GK Index · PYQ Blueprint.